# Ajay Venkatesh

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## Education

**New York University**, New York, NY — *Aug 2024 – May 2026*
*Master of Science in Computer Engineering* (GPA: 3.9/4.0)
*Coursework:* Machine Learning, Deep Learning, Big Data, System Design, Computer Architecture, Data Structures

## Experience

- **Campus Mesh** — *Jan 2026 – Present*
  *AI Software Engineer* — New York, US
  - Built **CampusIQ**, an LLM-powered AI assistant, with iterative **prompt engineering** and **RAG retrieval** over structured data and **embedding-based semantic search**, improving contextual accuracy by **30%**.
  - Designed a personalized **recommendation engine** using collaborative filtering on user interaction signals, increasing engagement by **28%** and retention by **50%**.
  - Developed scalable **Node.js microservices** powering real-time academic workflows for **10K+ users** across study groups, messaging, notifications, and profile management.
- **Amazon** — *May 2025 – Aug 2025*
  *Software Development Engineer Intern* — Seattle, US
  - Deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an **agentic workflow** that autonomously retrieves operational logs to reason over incident details, generating context-aware summaries and remediation steps.
  - Implemented a **Model Context Protocol (MCP)** server for efficient tool-augmented retrieval, cutting AWS Athena query time and compute costs by **40%**.
  - Built a high-throughput, event-driven incident triage system on **AWS ECS Fargate, SQS, SNS**, decoupling distributed message processing and reducing response latency by **50%**.
  - Provisioned resilient **Infrastructure-as-Code** using **AWS CDK** on auto-scaling ECS clusters, ensuring **99.9%** availability under 2x peak load.
- **New York University (Novel AI Technologies, Inc.)** — *Aug 2024 – Present*
  *AI Software Engineer* — New York, US
  - Architected an **NSF-funded ($100K)** AI-powered health platform for **iOS** and **Android** ingesting real-time wearable telemetry; implemented **time-series anomaly detection** using **Isolation Forest** and **LSTM** models.
  - Built end-to-end **computer vision pipelines** (**YOLOv8, EfficientNet**) for user-submitted food images, integrated with **GPT-4 vision workflows** for nutritional estimation and dietary feedback.
  - Developed a full-stack insurance analytics product (**React, Node.js, REST APIs**) integrating **LLM-based document analysis** (**GPT-4 + LangChain**) for underwriting insights.
  - Engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with control; processed **250K+** records and powered **AWS QuickSight** dashboards.
- **PricewaterhouseCoopers (PwC)** — *Aug 2021 – Aug 2024*
  *Senior Software Engineer* — Bengaluru, India
  - Designed a distributed backend on **Microsoft Azure** integrating with **Microsoft Dynamics** and supporting **ML inference pipelines** for healthcare analytics.
  - Built an **NLP-powered OCR pipeline** using **LayoutLM transformer** and Azure Form Recognizer to automate document data extraction, reducing processing time by **75%**.
  - Optimized data ingestion using **parallel processing** on **100K+** records, reducing execution from **5 hours to 1 hour**.
  - Led code reviews and mentored engineers on **software design and test-driven development**, reducing production bugs by **25%**.

## Projects

- **InviLite (AI-Driven Data Engineering & Analytics Platform)** | *AWS, SageMaker, Bedrock, FAISS, Python*
  - Architected a cloud-native data pipeline (Kinesis, Lambda, S3 Bronze/Silver/Gold, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed ML pipelines.
  - Built a **RAG system** (SVM classification + FAISS vector indexing + **Bedrock LLM**), cutting incident resolution from **1–2 hours to minutes** and lifting root-cause classification accuracy from **56% to 92%** via feature engineering and model retraining.
- **Hospital Management System** | *React, Node.js, Express, MongoDB, SageMaker, Hugging Face*
  - Built a full-stack healthcare platform (React, Node.js, Express, MongoDB) managing appointments, patient records, and lab reports with **role-based access control**.
  - Fine-tuned a **Hugging Face transformer-based model** reaching **88% accuracy** for health condition prediction and deployed on **AWS SageMaker** for real-time diagnostic support.

## Skills

- **Programming Languages:** Python, Java, C++, JavaScript (ES6), TypeScript, SQL
- **AI/ML:** PyTorch, TensorFlow, LLM Fine-tuning, RAG (LangChain, FAISS), Transformers, NLP, Hugging Face, Computer Vision (YOLOv8, EfficientNet), Time-Series Modeling, MLflow, SVM, Random Forest, Pandas, NumPy
- **Web & Backend:** React, Node.js, Express.js, REST API, GraphQL API, Microservices
- **Databases:** MySQL, PostgreSQL, MongoDB, DynamoDB
- **Cloud & DevOps:** AWS (Lambda, SageMaker, Bedrock, ECS, S3, CloudWatch, CDK, API Gateway, IAM), Azure (AI Services, Function Apps), Docker, Kubernetes, CI/CD, Terraform
- **Tools & Platforms:** Git, Linux, Jupyter, Postman, Cursor, GitHub Copilot, Model Context Protocol (MCP), Life Sciences Research, Biomedical Research, Scientific Computing, Genomics, Neuroscience, Drug Discovery, Context Engineering, Agent Architectures, Evaluation Frameworks, Production Systems Development

## Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques *(IEEE)*
An Aquila-optimized SVM Classifier for Diabetes Prediction *(IEEE)*

## Certifications

AWS Cloud Practitioner | Stanford Machine Learning (Coursera) | Azure Data Fundamentals